Snaptapp LLC — Google Cloud & Vertex AI Architecture Brief

Multi-Agent Gemini Inference, Google Search Grounding, Gemma 2 SFT & Cloud Run Infrastructure for **More or Less**

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1. AI-Native Game & Content Verification Engine Built on Google Cloud

Snaptapp LLC is an independent US studio building AI-native multiplayer trivia and social games on Google Cloud (`more-or-less-505908`). Our live title, More or Less, challenges up to 10 concurrent players per room to decide whether a real numerical fact is higher or lower than a calibrated distractor before a synchronized shot clock expires. Our core runtime, automated fact discovery and verification pipeline, real-time dispute referee, model fine-tuning workflow, and live-ops copilots are built natively on **Vertex AI** (`gemini-3.5-flash-lite`, `gemini-3.8-flash`, and fine-tuned `gemma-2-9b-it`), **Google Search Grounding**, **Google Cloud Run**, **Cloud Firestore**, and **Firebase**.

2. Production Google Cloud & Vertex AI Workloads (`more-or-less-505908`)

Subsystem	Google Cloud Stack	Model & Grounding	Production Role & Verification Mechanics
Live Pack Engine `agent-go/agent.go`	Cloud Run (Go 1.27), GenAI SDK `v1.73`, OpenTelemetry	`gemini-3.5-flash-lite` Vertex `global` + Search	Streams custom topic packs via NDJSON (`POST /pack/stream`) in <2s. Enforces mandatory per-round Google Search grounding (`grounding_receipt` telemetry) and deterministic **DDM-4** distractor calibration.
Fact Foundry & Auditor `foundry.go` · `factquality`	Cloud Run Jobs, Vertex AI, Cloud Firestore (`fact_bank`)	`gemini-3.8-flash` Vertex `global` + Search	Cold-path pipeline that crawls subjects, verifies numeric claims against primary web sources, classifies quantity kinds (`ResolveFactKind`), precomputes DDM-4 sweeps (surprise `0–3`), and gates `pending → active` promotion.
Fact Ombudsman `agent-go/arbitrate.go`	Cloud Run, Vertex AI, Cloud Firestore	`gemini-3.5-flash-lite` / `gemini-3.8-flash` + Search	Adversarial AI referee that investigates player fact disputes in real time via counter-evidence Google Search queries, issuing `DEBUNKED_AND_EVICTED`, `VERIFIED_CANONICAL`, or `AMBIGUOUS_UNIT` verdicts with cache eviction.
Vertex AI SFT & Gemma 2 `agent-go/sft_export.go`	Vertex AI Tuning, Gemma 2, Firestore (`model_adapters`)	Vertex Gemini SFT & `gemma-2-9b-it` LoRA	Hermetic exporter (`export-sft --format vertex`) compiles audited Firestore facts into Vertex AI tuning corpora and Gemma 2 9B candidate generators, gated by a 7-metric bar (`0.0%` redline and answer-leak rate).
Ops Portal AI Copilots ops.snaptapp.us	Vertex AI REST API, Cloud Run OIDC	`gemini-3.8-flash` Tool Calling & Search	Powers the interactive **Daily Blitz Strategist** (`daily-chat.server.ts`) for 15-round pack planning, the **Report Review Copilot** (`report-copilot.server.ts`) for dispute triage, and **Placement Insights** (`insights.server.ts`).
App Platform & Security `functions/` · `client/`	Firestore, Cloud Functions v2, Firebase Auth, Hosting, App Check	`@google/genai`, Play Integrity, App Attest, reCAPTCHA	Hosts the cross-platform Flutter Web client (WASM/Impeller), persists canonical profiles, leaderboards, and server-authoritative match histories (`recordPartyMatch`), and enforces device/token attestation.

3. End-to-End Hybrid Execution Pipeline & Credit Utilization Plan

- **Tiered Hot-Path Resolution (<1ms to <2s):** Covered topics serve from the compiled Firestore fact snapshot at the edge (<1ms) or warm cache (<10ms). Uncached custom topics pass a pre-quota safety and scope-purity gate before invoking `agent-go` on **Google Cloud Run** via OIDC, streaming Search-grounded **Vertex AI Gemini 3.5 Flash-Lite** rounds into synchronized WebSocket rooms.
- **Closed-Loop Data Flywheel & Deterministic Guardrails (DDM-4):** Models never invent distractor numbers or bypass verification. Every claim is validated by our deterministic mathematical kernel (`DDM-4`, shared across Go and TypeScript via SHA-256 digests), audited by **Vertex AI Gemini 3.8 Flash** in the Fact Foundry, and exported into **Vertex AI SFT** and **Gemma 2 9B** training sets.
- **24-Month Google Cloud Credit Utilization Roadmap:** Credits fund (1) scaling live **Vertex AI Gemini 3.5 Flash-Lite** inference and **Google Search Grounding** concurrency across Web, iOS, and Android launches; (2) expanding nightly **Vertex AI Gemini 3.8 Flash** Fact Foundry crawl, auditing, and dispute-arbitration throughput; (3) continuous **Vertex AI supervised fine-tuning** and **Gemma 2 9B** evaluation jobs; and (4) scaling **Cloud Run**, **Cloud Firestore**, and **Firebase** across *More or Less* and Snaptapp's second title in development.